

One Possible Worked Example

Fictional Core-route submission • not an answer key

Read this for the level of detail, not for answers to copy. The student, observations, screenshots, and numerical results below are invented. Your notebook will produce different evidence, and many strong responses will look different from this one.

This example models a useful pattern: record what you did, show the evidence you actually observed, and explain it in ordinary language.

CHECKPOINT 1

Getting oriented

EXAMPLE RESPONSE

I saved a copy named `HW1_Ada_Lovelace.ipynb` and added my name at the top. The runtime menu showed a T4 GPU. I changed the practice calculation from `2 + 2` to `7 * 6` and reran the cell, which replaced the old output with `42`.

Changing a cell does not automatically update later cells. I need to rerun the changed cell and any cells that depend on its result. After restarting the runtime, I need to run the notebook from the top because the saved variables are gone.

Why this works: it names the action, the visible result, and the general rule the student inferred.

CHECKPOINT 2

What did we train?

EXAMPLE RESPONSE

The inputs were pet photographs. The labels were "cat" and "not cat," derived from the file names. The model learned from a training portion of the images, while a separate validation portion was used to measure how well it handled examples it did not train on.

My fictional validation error rate was **0.024**, or about 2.4%. That means roughly 2 or 3 of every 100 held-out images were classified incorrectly in this run. It does not guarantee the same performance on every image found online.

Training changes the model using labeled examples. **Prediction** uses the trained model to produce an output for a new input without changing the model.

Why this works: it translates a metric into plain language and does not overclaim what the number proves.

CHECKPOINT 3

Inspect predictions

In a real submission, each record would include the actual image or notebook output.

Image A · clear cat portrait

Before running: cat, with high confidence. **Fictional output:** cat, 0.99.

The result matched my prediction. The face, ears, and whiskers were large and unobstructed, so the image resembled a straightforward training example.

Image B · small dog under a blanket

Before running: dog, but uncertain. **Fictional output:** cat, 0.63.

The model disagreed with me and was less confident. Most of the animal was hidden, and the visible pointed ears may have resembled features associated with cats. This is a useful failure because it suggests that context and occlusion matter.

Image C · fox photographed outdoors

Before running: neither label is appropriate. **Fictional output:** not cat, 0.91.

The binary classifier still had to choose one of its two labels. "Not cat" is technically compatible with a fox, but the output does not mean the model recognized a fox. The test exposes how the available labels limit what a prediction can mean.

Why these work: each record separates the student's prediction, the model's evidence, and the interpretation.

CHECKPOINT 4

Human-only reflection

No sample reflection is provided. This checkpoint is specifically reserved for your own 2–4 sentences. A model answer would work against the purpose of the question.

A useful reflection might begin from one concrete moment: a prediction that surprised you, a setup problem you resolved, or a question the comparison raised. Write the rest in your own words without AI drafting or polishing.

CHECKPOINT 5

Compare the examples

Pet classification · ran

Data: labeled pet photographs. Input: one image. Output: cat or not-cat probability. Model: a pretrained convolutional image classifier adapted to this task.

Image segmentation · inspected

Data: street-scene images with a category for each pixel. Input: one image. Output: a labeled region map. Model: a U-Net-style vision model.

Movie-review sentiment · inspected

Data: IMDB review text with positive or negative labels. Input: a review. Output: sentiment prediction. Model: a pretrained language model adapted for classification.

Adult-income tabular model · ran

Data: rows of demographic and employment fields. Input: one row of mixed categories and numbers. Output: an income category. Model: a tabular neural network.

Movie recommendation · ran

Data: users' movie ratings. Input: a user and movie pair. Output: a predicted rating. Model: collaborative filtering using learned user and movie factors.

EXAMPLE EXPLANATION OF HELD-OUT DATA

We hold out evaluation data because performance on the same examples used for training can be misleading. A validation set gives the model examples it did not learn from, so it is a better test of whether the learned pattern generalizes.

Example AI credit

I used Codex to explain the difference between validation error and training loss and to help interpret a file-path error. I wrote the notebook observations and reflection myself.

Fictional example drafted by Codex (OpenAI), based on the assignment and teaching direction of Xiuye Chen.